29/08/2026

A

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18,
Udyog Vihar, Gurugram, Haryana - 122015, ***Ph.***+91-1244168300

MM : 720

OYM - Term Test - 3
for NEET - 2028

Time : 180 Min.

Physics

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Chemistry

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Botany

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Zoology

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177. (3)
178. (2)
179. (1)
180. (4)

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Hints and Solutions

Physics

(1) **Answer :** (1)
Hint:
$\phi = \vec{B} . \vec{A}$.
Solution:
Magnetic field produced by coil having current will be perpendicular to area vector of other coil ϕ = *BA*cos90°
ϕ = 0.
Change in flux in other coil will be zero therefore no current will flow in other coil.

(2) **Answer :** (2)
Solution:
$U = -\vec{M} \cdot \vec{B}$
In stable equilibrium U = –*MB* = *x*
Torque when magnetic moment is perpendicular to the field = $\vec{M} \times \vec{B} = MB = \tau$
$\therefore \ \tau = -x \ \text{ or } \ \left|\frac{\tau}{x}\right| = 1$

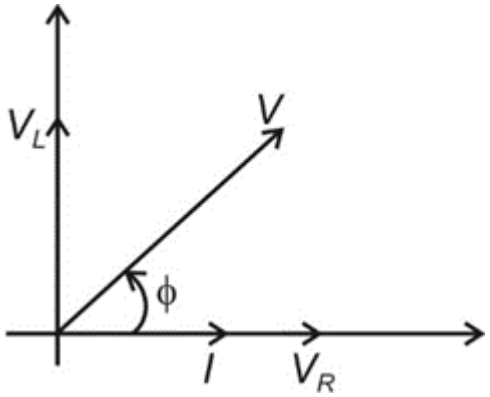
(3) **Answer :** (2)
Solution:
Starting from *B*, $V_B - L\frac{di}{dt} - 15 - IR = V_A$
$V_B - 5 \times 10^{-3}(-10^3) - 15 - 5 \times 1 = V_A$
⇒ $V_B - V_A$ = 15 V

(4) **Answer :** (4)
Solution:
Induced current in the circuit is zero as circuit is not complete. So magnetic force is zero.

(5) **Answer :** (3)
Solution:
$V^2 = V_R^2 + (V_L - V_C)^2$
Since $V_L = V_C$
Hence $V = V_R$ = 200 V

(6) **Answer :** (2)
Solution:
$\cos\phi = \frac{R}{Z} = \frac{40}{\sqrt{R^2 + (X_L - X_C)^2}}$
$= \frac{40}{\sqrt{1600 + 900}} = \frac{40}{50}$
$\boxed{\cos\phi = \frac{4}{5}}$

(7) **Answer :** (1)
Hint:
Voltage across inductor leads the current.
Solution:



$X_L = \omega L$

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= 200 × 100 × 10^{-3}
= 20 Ω

$\tan\phi = \frac{V_L}{V_R} = \frac{X_L}{R} = \frac{20}{20} = 1$

$\phi = \frac{\pi}{4}$

(8) **Answer :** (2)
Hint:
Use *KVL* and $e_L = -L\frac{di}{dt}$
Solution:
$V_P - 4 \times 4 - 10 - 5 \times 10^{-3} \times 10^4 - V_Q = 0$
$V_P - V_Q = 76$ V

(9) **Answer :** (1)
Solution:
To generate large current in inductance oil its resistance should be low.

(10) **Answer :** (4)

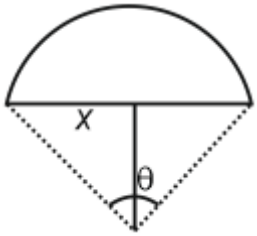
(11) **Answer :** (2)
Solution:
When magnet is cut perpendicular to length, time period of oscillation of each part
$T' = \frac{T}{n} = \frac{T}{2}$

(12) **Answer :** (2)
Solution:
Electric current is a scalar quantity.

(13) **Answer :** (1)
Solution:

$|\varepsilon| = \frac{d\phi}{dt} = 8t - 1$
$= 8 \times 2 - 1 = 15$ V

(14) **Answer :** (4)
Solution:



$\theta = \frac{arc}{R}$
Let length of magnet is *l* and Pole strength is *m*
then *M* = *ml*
$R = \frac{l}{\theta}$
When it bend in a are then
$M' = m(2x)$
$x = R\sin\frac{\theta}{2} = \frac{l}{\theta}\sin\left(\frac{\theta}{2}\right)$
$M' = \frac{2ml\sin}{\theta}\left(\frac{\theta}{2}\right)$
ml = *M*
$M' = \frac{2M\sin}{\theta}\left(\frac{\theta}{2}\right)$

(15) **Answer :** (4)
Solution:
$L = \mu_0 n^2 \pi r^2 l$
$A = \pi r^2$
$n = \frac{N}{l}$
$L = \mu_0 \frac{N^2}{l} A$

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By putting the given values, it can be seen that it is maximum for solenoid no. 4

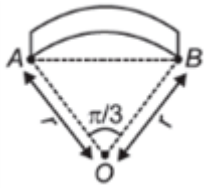
(16) **Answer :** (3)
Hint:
emf induced = $BA\omega\sin(\omega t)$
Solution:
$A = \frac{\pi r^2}{2}$
Emf induced = $e = B = \left(\frac{\pi r^2}{2}\right)$
$\omega\sin(\omega t)$
Average power $= \frac{e_0^2}{2R} = \frac{\pi^2 B^2 \omega^2 r^4}{8R}$

(17) **Answer :** (2)
Solution:
In purely inductive ac circuit, voltage leads current by phase 90°.

(18) **Answer :** (2)
Solution:
$e_{AB} = B[6\sin 30°]\ 10$
$= 3 \times 6 \times \frac{1}{2} \times 10 = 90\ \text{V}$

(19) **Answer :** (4)
Solution:
$\left|e_{\text{ind}}\right| = M\frac{di}{dt} = 15 \times 10^{-3}\frac{d}{dt}[5\sin(100\pi t + 45°)]\ \text{mV}$
$= 15 \times 10^{-3} \times 5 \times 100\pi\cos(100\pi t + 45°)\text{mV}$
$e_0 = 7.5\ \pi\text{mV}$

(20) **Answer :** (2)
Solution:



Distance between A and B = $AB = 2r\sin 30° = r$
New magnetic moment,
$M' = m \times r$
$= \frac{M}{l} \times \frac{3l}{\pi}$
$M' = \frac{3M}{\pi}$

(21) **Answer :** (1)

(22) **Answer :** (2)
Solution:
NCERT Reference:
Physics-XII, Page No.- 250

(23) **Answer :** (2)
Solution:

$V_o - V_C = \frac{\omega(2l)^2 B}{2} = 2\omega l^2 B$
$V_o - V_A = \frac{\omega l^2 B}{2}$
$\Rightarrow\ V_A - V_C = 2\omega l^2 B - \frac{\omega l^2 B}{2} = \frac{3\omega l^2 B}{2}$

(24) **Answer :** (2)
Solution:
$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{27\times10^{-3}\times30\times10^{-6}}}$
$= \frac{1}{9\times10^{-4}} = \frac{10000}{9} = 1111\ \text{rad/s}$

(25) **Answer :** (2)
Solution:
$Z = \sqrt{R^2 + X_L^2} = \sqrt{10^2 + 10^2} = 10\sqrt{2}\ \Omega$

(26) **Answer :** (2)
Solution:
$\phi = (5t^3 + 4t^2 + 2t - 5)$ Wb
time t = 2 s
Resistance = 5 Ω
Induced emf $E = \frac{d\phi}{dt}$
$\frac{d\phi}{dt} = 15t^2 + 8t + 2$
At $t = 2$ sec, $\frac{d\phi}{dt} = 15 \times 4 + 8 \times 2 + 2$
= 78 V
Also $E = IR$
E = 78 V, R = 5 Ω
$\frac{E}{R} = \frac{78}{5} = 15.6$ A

(27) **Answer :** (2)
Solution:
$L = \frac{\mu_0 N^2 A}{L}$
$L = \frac{4\pi \times 10^{-7} \times (400)^2 \times 0.2 \times 10^{-4}}{10 \times 10^{-2}}$
≈ 40 μH

(28) **Answer :** (2)

(29) **Answer :** (2)
Solution:
As an iron rod is inserted, inductance of inductor will increase. Hence impedance $z = \sqrt{R^2 + (\omega L)^2}$ will increase, so from $\mathrm{i} = \frac{\mathrm{V}}{\mathrm{Z}}$ current will decrease and from $P = \frac{V_0^2 R}{2Z^2}$ power of bulb will decrease.

(30) **Answer :** (3)
Hint:
At resonance $X_L = X_C$
Solution:
$\cos\phi = \frac{R}{Z}$
$Z = \sqrt{R^2 + (X_L - X_C)^2}$
$Z = R$
$\cos\phi = \frac{R}{R}$
= 1

(31) **Answer :** (1)
Solution:
$I = I_0 \sin(100\pi t)$
$I = \sin(100\pi t)$
$100\pi t = \frac{\pi}{2}$
$\boxed{t = \frac{1}{200}\, s}$

(32) **Answer :** (1)
Solution:
$\omega_0 = \frac{1}{\sqrt{LC}}$
$X_L = 2\omega_0 L$
$= 2\sqrt{\frac{L}{C}}$
$X_c = \frac{1}{2}\frac{1}{\omega_0 C}$
$= \frac{1}{2\sqrt{\frac{C}{L}}}$
$= \frac{1}{2}\sqrt{\frac{L}{C}}$
$\frac{X_L}{X_C} = 4 : 1$

(33) **Answer :** (1)
Solution:
$|\varepsilon_L| = L\frac{dI}{dt}$

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$\Rightarrow L = \frac{1.6}{2.4}$

$\boxed{L = \frac{2}{3}\text{H}}$

(34) **Answer :** (4)
Solution:
$I_{\text{rms}} = \frac{I_m}{\sqrt{2}}$

$\frac{I_{\text{rms}}}{I_m} = \frac{1}{\sqrt{2}}$

(35) **Answer :** (2)
Solution:
Resonance frequency
$f = \frac{1}{2\pi\sqrt{LC}}$

$f' = \frac{1}{2\pi\sqrt{L(4C)}} = \frac{f}{2}$

(36) **Answer :** (2)
Solution:
At resonance in LCR series a.c. circuit, voltage and current are in phase. Current will be maximum and impedance will be minimum at resonance.

(37) **Answer :** (1)
Solution:
An ideal choke coil is a pure inductor with no electrical resistance.
∴ Its power factor is exactly zero.

(38) **Answer :** (4)
Solution:
$Q = \frac{\Delta\phi}{R} \Rightarrow$ unit of $\frac{\Delta\phi}{R}$ is A s

volt = $\frac{\text{work done}}{\text{charge}} \Rightarrow$ charge = $\frac{\text{work done}}{\text{volt}}$

∴ unit of charge = J V^{-1}

(39) **Answer :** (3)
Solution:
$M = m \times l$
$\Rightarrow 4 = m \times 0.2$
$\Rightarrow m = \frac{40}{2} = 20 \text{ A m}$

(40) **Answer :** (4)
Solution:
$P = V_{\text{rms}} I_{\text{rms}} \cos \phi$
Phase difference between current and voltage in an inductor $\phi = 90°$
P = Zero

(41) **Answer :** (3)
Solution:
Transformers work for AC and are used to obtain a suitable AC voltage.

(42) **Answer :** (4)
Solution:
Between A and B inductors are in parallel
$\frac{1}{L} = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6}$
L = 2 H

(43) **Answer :** (4)
Solution:
The magnetic susceptibility is positive for paramagnetic and ferromagnetic material.

(44) **Answer :** (3)
Solution:
The magnetic susceptibility of a diamagnetic substance is negative and very slightly less than one and it is independent of temperature.

(45) **Answer :** (2)
Solution:

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An a.c. generator converts mechanical energy into electrical energy

Chemistry

(46) **Answer :** (4)
Solution:
Alkalies have no action on actinoids.

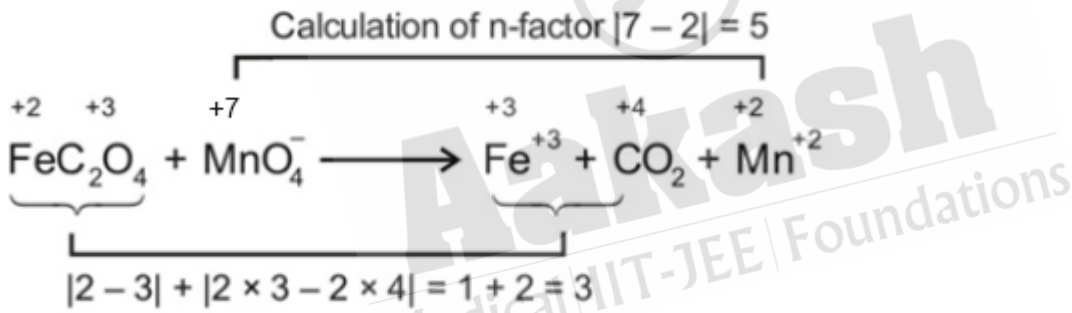
(47) **Answer :** (2)
Solution:
$\mu_s = \sqrt{n(n+2)}\ BM$
$Mn^{2+} = [Ar]3d^5 4s^0$
$n = 5$
$\mu = \sqrt{35}\ BM$

(48) **Answer :** (4)
Solution:
–SCN is an ambidentate ligand

(49) **Answer :** (2)
Solution:
EAN of Cr = 24 – 3 + 2 × 6 = 33

(50) **Answer :** (1)
Solution:
Stability of complex increases by chelation.

(51) **Answer :** (1)
Solution:

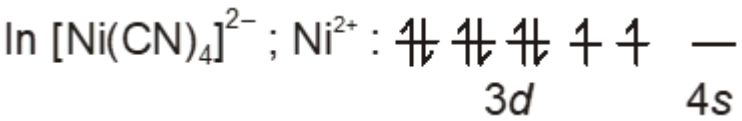


5 moles of FeC_2O_4 requires 3 moles of $KMnO_4$
1 mole of FeC_2O_4 requires $\frac{3}{5}$ moles of $KMnO_4$

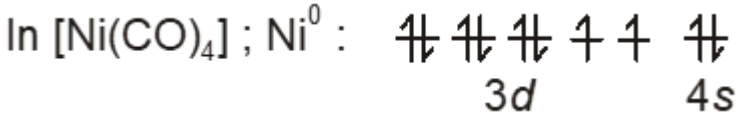
(52) **Answer :** (2)
Solution:
$d_{x^2-y^2}$ is used in square planar complexes.

(53) **Answer :** (4)
Solution:
MnO_4^- is stronger oxidising agent than $Cr_2O_7^{-2}$

(54) **Answer :** (3)
Solution:



CN^- being a strong-field ligand pairs up the two 3*d* electrons; one 3*d* - orbital becomes available and the hybridization is dsp^2.



Again, CO being a strong-field ligand pairs up the two 4*s* electrons with 3*d* electrons; the hybridisation is sp^3.

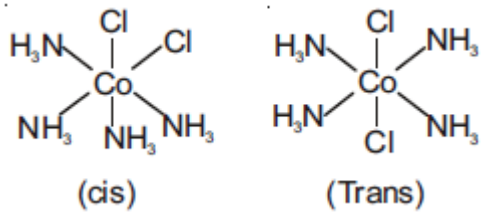
(55) **Answer :** (1)
Hint:
I_2 is a weaker oxidising agent.
Solution:
Cu^{2+} can oxidise I^- to I_2. That's why CuI_2 does not exist.

(56) **Answer :** (3)
Solution:
Basic strength of hydroxide of lanthanoids decreases from $Ce(OH)_3$ to $Lu(OH)_3$.

(57) **Answer :** (3)
Solution:
In contact process, V_2O_5 is used as a catalyst.

(58) **Answer :** (3)
Solution:
Nd(Z = 60) → [Xe] $4f^4 6s^2$
Number of unpaired electrons = 4

(59) **Answer :** (2)
Solution:



(60) **Answer :** (1)
Solution:
Ni^{2+} and Ti^{3+} both have partially filled *d*-subshell therefore are coloured in aqueous solution.

(61) **Answer :** (4)

(62) **Answer :** (2)
Solution:
Os shows +8 oxidation state.

(63) **Answer :** (1)
Solution:
Ma_3b_3 type compound will have two geometrical and zero optical isomerism.

(64) **Answer :** (3)
Solution:
The valence shell electronic configuration of given species is as follows

Nd^{3+}	$4f^3$	Three unpaired electrons
Gd^{3+}	$4f^7$	Seven unpaired electrons
Lu^{3+}	$4f^{14}$	Zero unpaired electron
Eu^{2+}	$4f^7$	Seven unpaired electrons

(65) **Answer :** (4)
Solution:
$4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \rightarrow 8Na_2CrO_4 + 2Fe_2O_3 + 8CO_2$

(66) **Answer :** (3)
Solution:
$[Mn(CN)_6]^{3-}$ has magnetic moment of two unpaired electrons and it is inner orbital complex involving d^2sp^3 hybridization.

(67) **Answer :** (1)
Solution:
$[Co(CN)_6]^{3-}$ has Co^{3+} *i.e.*, Co^{3+} = [Ar] $d^6 4s^\circ$, CN^- ion is a strong field ligand forms a low spin complex with central metal ion. So there is no unpaired electron in this complex.

(68) **Answer :** (4)
Solution:

All the elements after uranium$_{(Z = 92)}$ are transuranic elements.

(69) **Answer :** (3)

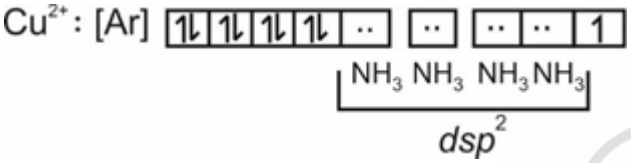
(70) **Answer :** (2)

(71) **Answer :** (2)
Solution:
Iodide ion is oxidised to iodate ion
$2\,MnO_4^- + H_2O + I^- \rightarrow 2\,MnO_2 + 2\,OH^- + IO_3^-$

(72) **Answer :** (2)
Solution:
$Mn^{2+} = {}_{18}[Ar]3d^5$
$\therefore$ Number of unpaired electrons = 5 (highest out of given species)
$\Rightarrow \mu$ (magnetic moment) $= \sqrt{35}\ BM$

(73) **Answer :** (3)
Solution:

Element	Ce	Pm	Eu	Gd
Radii (pm)	183	181	199	180

(74) **Answer :** (2)
Solution:


(75) **Answer :** (3)

(76) **Answer :** (3)

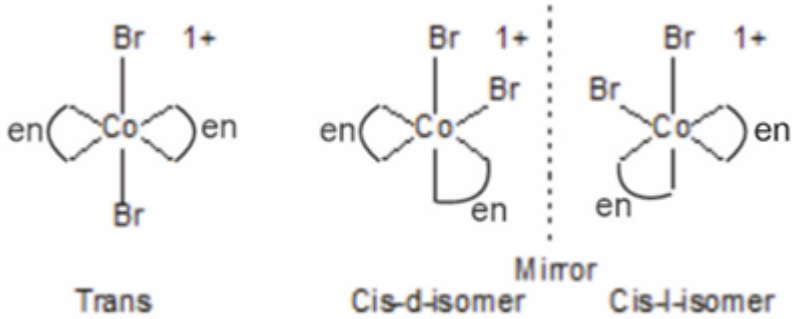
(77) **Answer :** (1)
Solution:
Strong 'Jahn-Teller' effect observed in incomplete e_g orbitals.

(78) **Answer :** (4)
Solution:
Stable fluorides of Mn are MnF_2, MnF_3 and MnF_4.

(79) **Answer :** (2)
Solution:
Due to stable electronic configuration, the melting point of Mn is lowest in the given elements.

(80) **Answer :** (1)
Solution:
Configure of element will be $[Ar]3d^34s^2$, which belongs to V atomic number 23.

(81) **Answer :** (3)
Solution:
Ce^{4+} and Yb^{2+} acquire $[Xe]4f^0$ and $[Xe]4f^{14}$ electronic configurations, respectively.

(82) **Answer :** (2)
Solution:


(83) **Answer :** (2)
Solution:
More number of ligands

(84) **Answer :** (3)
Solution:
Enthalphy (in kJ mol^{-1}) of atomisation :
Ti → 473
Cr → 397
Mn → 281
Co → 425

(85) **Answer :** (3)

(86) **Answer :** (1)
Solution:
The order of the strength of ligands in the spectrochemical series is
$H_2O < NH_3 < en$
Hence, the wavelength absorbed will be in the order
$[Co(H_2O)_6]^{3+} > [Co(NH_3)_6]^{3+} > [Co(en)_3]^{3+}$

(87) **Answer :** (4)
Solution:
$NO_2^- \rightarrow NO_3^- + 2e^-$
$MnO_4^- + 5e^- \rightarrow Mn^{2+}$
So, moles of $KMnO_4 = \frac{2}{5}$

(88) **Answer :** (1)

(89) **Answer :** (3)
Solution:
The secondary valency or CN of cobalt in $CoCl_3.6NH_3$, $CoCl_3.5NH_3$ and $CoCl_3.4NH_3$ is equal to 6.

(90) **Answer :** (3)
Solution:
More the number of unpaired electrons (n), more will be magnetic moment.
$[Mn(CN)_6]^{4-}$, n = 1
$[Cr(NH_3)_6]^{3+}$, n = 3
$[FeF_6]^{3-}$, n = 5
$[Cu(NH_3)_4]^{2+}$, n = 1

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Botany

(91) **Answer :** (4)
Solution:
Tightly linked genes show less recombination frequency and thus less individuals in the offspring with non-parental gene combination.

(92) **Answer :** (3)
Solution:
Phenylketonuria is an inborn error of metabolism.
The affected individuals lack enzyme phenylalanine hydroxylase that converts phenylalanine into tyrosine. As a result of this, phenylalanine is accumulated and converted into phenylpyruvic acid and other derivatives.

(93) **Answer :** (2)
Solution:
Chromosomal theory of inheritance was proposed independently by Sutton and Boveri and experimentally verified by Morgan.

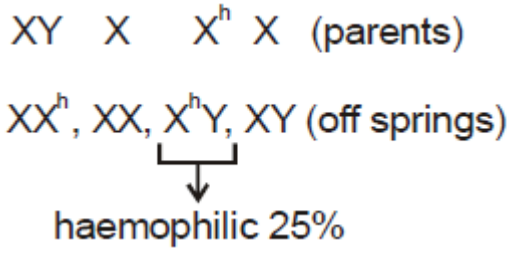
(94) **Answer :** (3)
Solution:
ZW-ZZ type of sex determination involve female heterogamety.
ZW-ZZ type of sex determination is seen in birds.

(95) **Answer :** (2)
Solution:
This pedigree chart is true for both X-linked recessive and autosomal recessive inheritance.

(96) **Answer :** (3)
Solution:

Down's syndrome is a chromosomal disorder.

(97) **Answer :** (3)
Solution:



(98) **Answer :** (1)
Solution:
(A) 2n – 1 → Monosomic condition
(B) 2n – 2 → Nullisomic condition
(C) 2n + 1 → Trisomic condition
(D) 2n + 2 → Tetrasomic condition

(99) **Answer :** (3)
Solution:
Turner's syndrome is a disorder caused due to the absence of one of the X chromosomes, *i.e*., 45 with X0, such females are sterile, as ovaries are rudimentary besides other features including lack of other secondary sexual characters.

(100) **Answer :** (1)
Solution:
Thalassemia and sickle cell anaemia are similar as the genes responsible for both of them are present on chromosome 11.

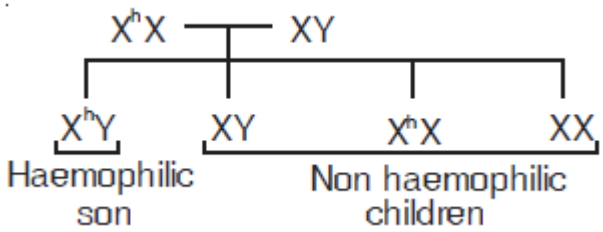
(101) **Answer :** (3)
Hint:
More parental type offsprings show less recombination frequency.
Solution:
In the second experiment, the recombinant types are more as compared to the first experiment. It means, cross over value (recombination frequency) for eye colour and wing type is more than that of body colour and eye colour.

(102) **Answer :** (1)
Solution:
In sickle cell anaemia, the mutant haemoglobin molecule undergoes polymerization under low oxygen tension causing change in the shape of the RBC from biconcave disc to elongated sickle like structure.

(103) **Answer :** (3)
Solution:
During linkage the proportion of parental gene combination is much higher

(104) **Answer :** (3)
Solution:
A.H. Sturtevant on the basis of recombination frequency, prepared first chromosomal map for *Drosophila*.

(105) **Answer :** (3)
Hint:
Haemophilia is an X-linked recessive disorder.
Solution:
Male – Non-haemophilic (XY)
Female – Non-haemophilic XX or X^hX



Thus, if a woman is carrier (normal) and man is normal, they can have a haemophilic son.

(106) **Answer :** (4)
Solution:
Colourblindness shows criss cross inheritance as it is a X-linked recessive disorder. For a daughter to be colour-blind, she must inherit the recessive allele from both parents.

A colour-blind man cannot pass the disorder to his sons because sons inherit the Y chromosome from their father, not the X chromosome. However, he can pass the X chromosome carrying the disorder to his daughters, who can then pass it to their sons (his grandsons).

(107) Answer : (3)
Solution:
aa × y^h
$ay^h = 1 \times \frac{1}{2} = \frac{1}{2}$

(108) Answer : (2)
Solution:
Male honey bees are haploid individuals and they produce gametes by mitosis.

(109) Answer : (1)
Solution:
Complete linkage no crossing over, off springs like parents.

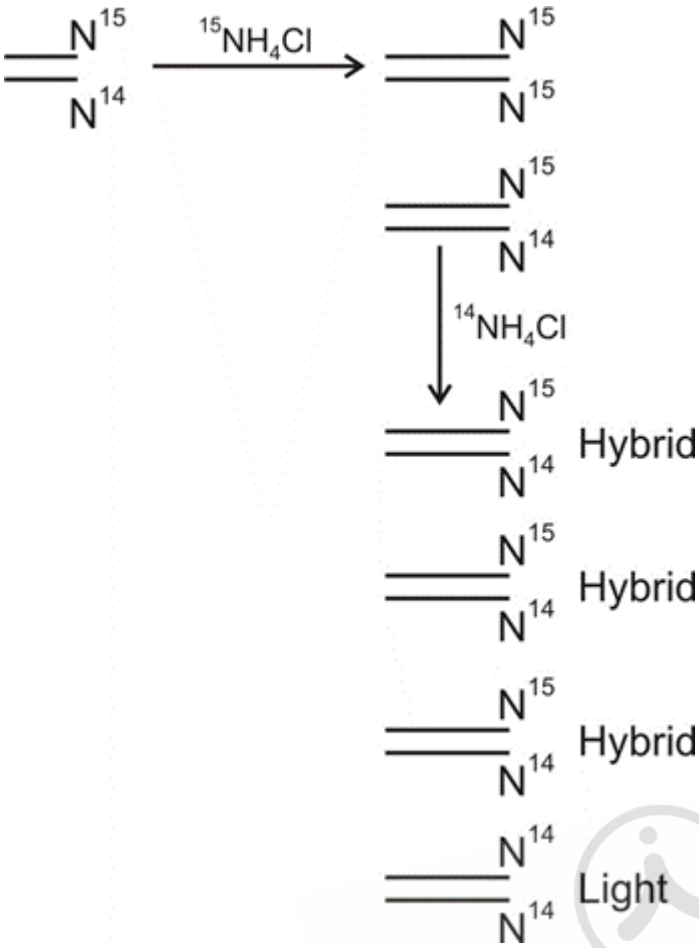
(110) Answer : (3)
Hint:
In dsDNA molecule
Purine = Pyrimidine
i.e., A = T, C = G
Solution:
C + G $\rightarrow$ 28 + 28 = 56%
Then A + T $\rightarrow$ 44%
A or $T = \frac{44}{2} = 22\%$

(111) Answer : (1)
Solution:
On the template strand of DNA with polarity 5' → 3' DNA synthesis is discontinuous in the form of Okazaki fragments. These fragments are joined by the enzyme DNA ligase.

(112) Answer : (3)
Solution:
Two nucleotides are linked through 3′–5′ phosphodiester linkage to form a dinucleotide. In a nucleotide 5′C have phosphate group and 3′C have hydroxyl group.

(113) Answer : (1)
Solution:
DNA shows semi-conservative mode of replication.

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Heavy = 0%
Hybrid = 75%
Light = 25%

(114) Answer : (4)
Hint:
Two terminal phosphates in a deoxyribonucleoside triphosphates are high energy phosphates.
Solution:
In addition to acting as substrate, deoxyribonucleoside triphosphates also provide energy for polymerisation.

(115) Answer : (2)
Solution:
1 nucleosome = 200 bp
$= \frac{7.2\times10^{6}\text{bp}}{200\text{bp}}$
No. of nucleosomes = 36×10^3

(116) Answer : (4)
Solution:
There is a definite region in *E. coli* DNA, where the replication originates.
Such regions are termed as origin of replication.

(117) Answer : (3)
Solution:
Helicase unwinds DNA helix during DNA replication.

(118) Answer : (2)
Solution:
DNA is less reactive and more stable hence preferred as genetic material.

(119) Answer : (4)
Solution:
Stacking of one base pair over the other in double helix, hydrogen bonding and presence of thymine instead of uracil, all confer additional stability to DNA

(120) Answer : (4)
Solution:
According to Chargaff's rule $\frac{\text{A}+\text{G}}{\text{T}+\text{C}} = 1$

Bacteria have higher G + C content, thus A+T/G+C is less than unity.

(121) Answer : (4)
Solution:
In the double helix structure of B-DNA, the pitch of the helix is approximately 3.4 nm (or 34 Å), and there are roughly 10 base pairs (bp) in each turn.

(122) Answer : (4)
Solution:
The promoter is said to be located towards 5'-end (upstream) of the structural gene (the reference is made with respect to the polarity of coding strand). The terminator is located towards 3'-end (downstream) of the coding strand.

(123) Answer : (3)
Solution:
$\phi \times 174$ has 5386 nucleotides.

(124) Answer : (2)
Solution:
'Reverse of Central Dogma of Biology' is also known as Teminism.

(125) Answer : (3)
Hint:
One chain in dsDNA has the polarity 5′ → 3′, the other has 3′ → 5′.
Solution:
dsDNA is made of two polynucleotide chains, where the backbone is constituted by sugar - phosphate, and the bases are projected inside. The two chains have anti-parallel polarity. Purine comes opposite to a pyrimidine with the help of hydrogen bond and this generates approximately uniform distance between the two strands of the helix. The two chains are coiled in a right-handed fashion.

(126) Answer : (2)
Solution:
H_1 is not part of nucleosome

(127) Answer : (4)

(128) Answer : (2)

(129) Answer : (4)
Solution:
DNA replication in prokaryotes is catalysed by DNA dependent DNA polymerase.

(130) Answer : (1)
Solution:
Helicase catalyses breaking of H bonds and separates two DNA strands.

(131) Answer : (2)
Solution:
RNA of viruses can directly synthesize proteins.

(132) Answer : (3)
Solution:

– Affected female

(133) Answer : (3)
Solution:
Friedrich Meischer named the acidic substance present in nucleus as Nuclein.

(134) Answer : (3)
Solution:
Heterochromatin region have highly condensed and tightly packed DNA.
Heterochromatin is genetically inactive and does not participate in transcription, while euchromatin is genetically active.

(135) Answer : (3)
Solution:
Gynaecomastia is a symptom shown by individual inflicted with Klinefelter's syndrome.

Zoology

(136) Answer : (1)

Solution:
GnRH is a releasing hormone responsible for the release of gonadotropins *i.e*., FSH and LH from the anterior pituitary gland.

(137) Answer : (3)
Solution:

ECF contains 30 times less K^+ than axoplasm while ECF contains 10 times more Na^+ than ICF. Potential difference around -70 mV exists across axolemma due to differential distribution of ions. Neural membrane is impermeable to movement of negatively charged proteins. The exterior surface of axolemma is positively charged.

(138) Answer : (2)
Solution:

PTH : Parathyroid hormone (peptide hormone) increases level of Ca^{+2} in blood (hypercalcemic).
It acts on bones and stimulates the process of bone resorption.

Also stimulates reabsorption of Ca^{+2} by renal tubules and from digested food.

(139) Answer : (4)
Solution:
Glucagon increases glucose level in the plasma from carbohydrate or non carbohydrate sources.

(140) Answer : (4)
Solution:
During deficiency of glucose in body cells, they start oxidising fatty acids to generate ketone bodies, which are excreted along with urine *i.e*., Ketonuria. Ketonuria and glycosuria (presence of glucose in urine) are symptoms of diabetes mellitus which is caused to deficiency of insulin.

(141) Answer : (2)
Solution:
Iter is the duct that connects third and fourth ventricles.
Lateral ventricles are called paracoel. Third ventricle is called diocoel.

(142) Answer : (1)
Solution:
Melatonin is secreted by pineal gland located on diencephalon.

(143) Answer : (3)
Solution:
Oxytocin acts on smooth muscles whereas vasopressin on renal tubule

(144) Answer : (4)
Solution:
Secondary sexual characters are maintained by estrogen in females and testosterone in males.
Androgen secretion from testes is stimulated by LH. Maintenance of corpus luteum and ovulation is also under control of LH. LH and FSH are gonadotropins.

(145) Answer : (2)
Solution:
Addison's disease occurs due to hyposecretion of hormones secreting from adrenal cortex.

(146) Answer : (1)
Solution:
Presence of plasma membrane is feature of all animal cells.

(147) Answer : (4)
Solution:
Epinephrine, cortisol, insulin and glucagon influence the carbohydrate metabolism

• Aldosterone is the main mineralocorticoid which acts on renal tubules and stimulates the reabsorption of Na^+ and water.

(148) Answer : (3)
Solution:
Melatonin plays a very important role in the regulation of biological clock. Melanin is a skin pigment.
MSH is a melanocyte stimulating hormone which acts on the melanocytes and regulate pigmentation of the skin.

(149) Answer : (4)
Hint:
Identify the hormone which proliferates alveoli in mammary glands.
Solution:

Both parathormone and thyrocalcitonin are antagonistic hormones and regulate blood Ca^{+2} level. Source structure of ANF is heart.

(150) Answer : (3)
Solution:

Hormones are defined as non-nutrient chemicals which act as intercellular messengers that are produced in trace amounts.

(151) Answer : (4)
Hint:
Cortisol and histamine are antagonistic.
Solution:
Cortisol acts as life saving hormone and is anti-inflammatory in response.

(152) Answer : (3)
Solution:
Steroidal hormones can directly act on DNA through intracellular receptors.

(153) Answer : (2)
Solution:
Dwarfism is caused due to hyposecretion of growth hormone.

(154) Answer : (3)
Hint:
Hyperpolarised neurons have potential lesser than resting membrane potential.
Solution:
–70 mV is the general value of resting membrane potential for a neuron. –55 mV can be the value for threshold potential when a neuron is stimulated.

(155) Answer : (3)
Hint:
It induces sleep.
Solution:
Pineal gland regulates diurnal rhythm by secreting melatonin.
Light inhibits melatonin production in body. Melatonin levels spike at night.

(156) Answer : (3)
Solution:
Medulla oblongata is the gastric secretion control centre. Pneumotaxic centre is present in pons.

(157) Answer : (4)
Solution:
Bone marrow is site of action of erythropoietin.

(158) Answer : (4)
Solution:
Both glucagon and insulin are proteinaceous in nature and pancreas is the source gland for both the hormones.

(159) Answer : (3)
Solution:
Sympathetic nervous system regulates the activity of body during emergency conditions leading to increase in heart rate, dilation of pupil, inhibition of salivation etc.

(160) Answer : (2)
Solution:
Hypothalamus secretes releasing hormones which in portal veins reach adenolypophysis.

(161) Answer : (2)
Solution:
Cerebral aqueduct is a canal which passes through midbrain and connects 3rd ventricle of diencephalon with 4th ventricle of medulla oblongata.

(162) Answer : (4)
Solution:
Resting membrane potential is maintained by the sodium-potassium pump.
Neurons are excitable cells because their membranes are in a polarised state. The electrical potential difference across resting plasma membrane is called the resting potential. The potential difference across the plasma membrane in depolarised state is called action potential, which is in fact termed as nerve impulse.

(163) Answer : (2)
Hint:
Aldosterone is a salt retaining hormone.
Solution:
Aldosterone increases reabsorption of Na^+ and Cl^- from filtrate and promotes secretion of K^+ and phosphate ions. So, in case of tumor formation, increased release of aldosterone will lead to decrease in Na^+ but increase in K^+ in urine.

(164) Answer : (2)
Hint:
Deficiency of calcium is associated with this disease.

Solution:
Low PTH (parathormone) leads to deficiency of calcium levels in blood and results in tetany.

(165) Answer : (2)
Hint:
This gland stores hormones in Herring's bodies.
Solution:
Hormones which act on the endocrine system but are synthesized within neurosecretory cells or neurons are called neurogenic hormones. Hypothalamus and adrenal medulla secrete neurogenic hormones but neurohypophysis only stores and releases neurogenic hormones. Thyroid gland stores its hormones in an inactive form but they are not neurogenic.

(166) Answer : (4)
Solution:
FSH acts on cells of Sertoli to secrete some factors which help in spermiogenesis

(167) Answer : (3)
Solution:
ANF can cause vasodilation and thereby decrease the blood pressure. It is antagonistic to renin-angiotensin aldosterone system.

(168) Answer : (3)
Hint:
It maintains circadian rhythm.
Solution:
Melatonin is derived from amino acid tryptophan.

(169) Answer : (2)
Solution:
Two cerebellar hemispheres are connected together by vermis. Colliculi are present in midbrain.

(170) Answer : (3)
Hint:
Catabolic reaction
Solution:
Insulin from β-cells lowers blood glucose level by promoting glycogenesis.

(171) Answer : (1)
Hint:
Disease also called exopthalmic goitre
Solution:
Grave's disease or exopthalmic goitre is caused due to hyperthyroidism. This disease is characterised by protrusion or bulging of eyeballs. Cretinism, Myxedema/ Gull's disease and simple goitre are caused due to hypothyroidism.

(172) Answer : (4)
Solution:
Medulla oblongata controls respiration, cardiovascular reflexes and gastric secretions.

(173) Answer : (3)
Hint:
Role of gap junctions.
Solution:
The membranes of pre and post-synaptic neurons are in very close proximity. Pre and post synaptic neurons are connected together by connexons.

(174) Answer : (3)
Solution:
ANF and ADH are antagonistic to each other w.r.t blood pressure.

(175) Answer : (3)
Hint:
This pump maintains the electrochemical gradient between ICF and ECF.
Solution:
In resting state, voltage gated Na^+ and K^+ channels are closed. Sodium-potassium pumps require energy to transport $3Na^+$ outwards for $2K^+$ into the axoplasm because sodium-potassium pump works against the concentration gradient of Na^+ and K^+, whereas the Na^+ and K^+ leaky ion channels transport ions from high to low concentrations.

(176) Answer : (1)
Solution:
Nissl's granules are present in cyton and dendrites of a neuron.

(177) Answer : (3)
Hint:
Part of brain containing optic lobes.

Solution:
Dorsal portion of midbrain consists mainly of four round swellings which are named as optic lobes/colliculi. All these swellings are collectively called corpora quadrigemina.

(178) Answer : (2)
Hint:
It also acts on liver to secrete bile juice.
Solution:
Secretin is a proteinaceous hormone synthesized from duodenal mucosa and acts on liver as well as exocrine part of pancreas. Released pancreatic juice contains water and sodium bicarbonate to neutralise gastric HCl. CCK also acts on pancreas to secrete pancreatic enzymes. Gastrin increases secretion of gastric juice and GIP is antagonistic to gastrin.

(179) Answer : (1)
Solution:
Presence of myelin sheath on nerve fibres (axon) gives a white appearance.
The neuronal cell bodies are concentrated in the cerebral cortex and hence it appears grey in colour. The myelinated axons occur in the inner part of cerebrum which is called white matter.

(180) Answer : (4)
Solution:
Somatotropin is growth hormone and is secreted by anterior pituitary. Other hormones are released by hypothalamus.

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